



Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Transfer Learning for Imbalanced Datasets

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Received: 17 August 2024 / Revised: 3 November 2024 / Accepted: 6 November 2024 / Published online: 18 November 2024
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Abstract

Pneumonia remains a significant global health challenge, necessitating timely and accurate diagnosis for effective treatment. In recent years, deep learning techniques have emerged as powerful tools for automating pneumonia detection from chest X-ray images. This paper provides a comprehensive investigation into the application of deep learning for pneumonia detection, with an emphasis on overcoming the challenges posed by imbalanced datasets. The study evaluates the performance of various deep learning architectures, including visual geometry group (VGG), residual networks (ResNet), and Vision Transformers (ViT) along with strategies to mitigate the impact of imbalanced dataset, on publicly available datasets such as the Chest X-Ray Images (Pneumonia) dataset, BRAX dataset, and CheXpert dataset. Additionally, transfer learning from pre-trained models, such as ImageNet, is investigated to leverage prior knowledge for improved performance on pneumonia detection tasks. Our investigation extends to zero-shot and few-shot learning experiments on different geographical regions. The study also explores semi-supervised learning methods, including the Mean Teacher algorithm, to utilize unlabeled data effectively. Experimental results demonstrate the efficacy of transfer learning, data augmentation, and balanced weight in addressing imbalanced datasets, leading to improved accuracy and performance in pneumonia detection. Our findings emphasize the importance of selecting appropriate strategies based on dataset characteristics, with semi-supervised learning showing particular promise in leveraging unlabeled data. The findings highlight the potential of deep learning techniques in revolutionizing pneumonia diagnosis and treatment, paving the way for more efficient and accurate clinical workflows in the future.

Keywords X-ray images · Visual geometry group (VGG) · Residual networks (ResNet) · Vision Transformers (ViT) · ImageNet · Deep learning

Introduction

Pneumonia is a lung infection caused by bacteria, viruses, or fungi, and it takes place when an infection causes the alveoli, or lung air sacs, to swell up with fluid or pus. As a result of lung inflammation, individuals with pneumonia experience difficulty breathing in sufficient oxygen for their bloodstream. According to the World Health Organization

(WHO), vulnerable populations with weakened immune systems, such as children under 5 and seniors over 65, are particularly susceptible to pneumonia, which is the leading cause of death among children under 5, accounting for more than one million deaths worldwide annually [1]. Given the severity of this disease, early and accurate detection of pneumonia is crucial to facilitate prompt treatment and management and to reduce its public health impact. To this end, the potential impact of machine learning in biomedical image analysis has generated significant interest in developing an automated, accurate, and robust framework to detect pneumonia from chest X-ray images to improve the accuracy and efficiency of pneumonia diagnosis, particularly in resource-limited settings where access to experienced radiologists may be limited.

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Machine learning models have been slowly gaining traction with medical data analysis following their successes in the field supported by a very high accuracy with their predictions. Most datasets that are used for the training of these models are ensured to have a mixed diversity with their datasets, but the diversity is usually never in an even distribution [2]. Disparities in healthcare data among underrepresented regions can result in the development of suboptimal AI models with poor prediction accuracy and robustness, potentially causing significant negative effects on healthcare for these underrepresented regions. This idea was explored via a paper published in 2020 highlighting these disparities. The paper decided to specifically focus on cancer genomics databases and saw that the data collected was primarily from Caucasians (91.1%), distantly followed by Asians (5.6%), African Americans (1.7%), Hispanics (0.5%), and other populations (0.5%) [3]. This meant that if all of the data was to be collated and used uniformly as part of the training data, then the predictions made using information from those databases would be more accurate on the well-represented Caucasian population as opposed to the others that were severely underrepresented. The paper in question took a look at this particular theory and proved it using a series of learning schemas: mixture learning, independent learning, and transfer learning. Mixture learning was done by collating all of the data together as previously mentioned. Independent learning was done by attempting to run all of the population data separately, building unique models for each population. While this would in fact provide for us tailor-made models that would have very high accuracy for each population, the issue lies there in the lack of data present for the underrepresented population. Transfer learning solves the issues with mixture learning and independent learning. It still divides up the population and trains a separate model for each one; however, it uses the knowledge obtained from the largest population to develop the other models. With a starting point provided for the models needed for each of the other datasets, it takes much less time and data to optimize each model to fit its respective population. Transfer learning ended up providing higher prediction accuracy with the underrepresented regions establishing a precedence to use transfer learning with similar situations in regard to medical data. The application of this idea to different types of medical data such as image data is not something that has been fully explored yet, especially if geographical data is pulled into the mix as a potential variable.

Main Contribution and Novelty

Our proposal aims to significantly contribute to the domain of biomedical image investigation by exploiting transfer learning to use on X-ray image datasets obtained from different geographical datasets. Specifically, we will focus on

two different datasets from distinct geographical regions, a larger dataset from Guangzhou, China, with 5856 X-ray images taken from the anteroposterior position (AP) and a smaller dataset from the Hospital Israelita Albert Einstein in Brazil, with 170 images. One of our primary contributions is the investigation of the effect of using geospatial information as a classifying variable in predictions with datasets from different geographical regions. We intend to study whether geospatial should be considered a relevant element when making predictions using biomedical image databases, given that it has been shown to affect diagnosis accuracy in other medical applications. In addition, the significance of transfer learning with underrepresented geographical regions in biological image analysis will be determined. By focusing on datasets with smaller populations, we want to highlight the potential advantages of transfer learning for enhancing performance with underrepresented regions.

Last but not least, we will generate extra images from existing X-ray images to balance our dataset, allowing for more precise deep-learning model training. By tackling the issue of imbalanced datasets, this unique approach has the potential to increase the accuracy of biological image analysis models. These innovative contributions can potentially advance biomedical image analysis as a whole considerably.

In this research work, six sections are established as follows: In the “[Introduction](#)” section, the utilization of deep learning concerning medical imaging is discussed. The “[Literature Review](#)” section comprises a literature review where we examine related articles, their datasets including Chest X-Ray Images (Pneumonia), and the results derived from them. In the “[Methodology](#)” section, the methodology examines the datasets used in the study, which include Chest X-Ray Images (Pneumonia) dataset, the BRAX dataset, and the CheXpert dataset. The findings of the experiment are analyzed and explained in the “[Results](#)” section. The highlights and limitations of the study are discussed in the “[Discussion](#)” section. The “[Conclusion](#)” section concludes the research paper, while the “[Future Work](#)” section explores future work related to this study.

Literature Review

Chest X-rays are a familiar and valuable tool in medical diagnosis and treatment. However, without a qualified radiologist, it is challenging to interpret. Therefore, many scientists are working to automate X-ray analysis. In recent years, the use of machine learning models including deep learning and transfer learning in the medical field has been growing, with clear successes in the predictions of various medical conditions. However, most datasets used for training these models are not diverse enough and this leads to disparities in healthcare data across different geographical

regions. Several studies have been conducted to address the issue of data disparities in medical image analysis. This issue was highlighted in a 2020 study which found that data collected for cancer genomics databases was primarily from Caucasians (91.1%), followed by Asians (5.6%), African Americans (1.7%), Hispanics (0.5%), and other populations (0.5%) [3].

A study by Park et al. (2023) [4] proposed a transfer learning approach to address the issue of data scarcity in the analysis of chest X-ray images for tuberculosis (TB) diagnosis. The study showed that transfer learning improved the performance of the deep learning model, compared to training the model from scratch. Another study conducted in February 2020 by Gaurav Labhane et al. [5], developed four neural network models to detect pneumonia from the chest X-ray images using CNN and transfer learning methodologies. These models consist of a basic convolutional neural network (CNN), VGG16, VGG19, and InceptionV3. Models were trained on 2992 pneumonia and 2972 normal chest X-rays and tested on 854 pneumonia and 849 normal. These models achieved accuracy over 97% [6].

In June 2022, a paper was published that saw that AI models being created for medical image analysis considered small biological differences between patients including age, sex, and geographical origin [7]. This result was extremely surprising considering that up until now most radiologists considered medical images to be racially agnostic. With the possibility that machine learning could possibly detect geographical patterns from these images in play, there arises the possibility of utilizing this information for increased performance. Demographics, such as geography, play a role in disease risk and outcome as there are clear differences in disease prevalence between different regions, and the biases that crop up from doing simple mixture learning for predictions could mask these factors leading to poorer prediction accuracies for less well-represented populations exactly as we saw in the 2023 paper [8]. Applying the findings of that paper, which addressed data disparities through transfer learning, to geospatial medical image datasets could result in higher accuracy results in disease outcome predictions than can be obtained using current machine learning methods.

Both supervised and unsupervised approaches may fall short of the intended outcome in situations when there are insufficient labeled data or where clustering is challenging. Semi-supervised learning [9] is a broad class of machine learning techniques that use both labeled and unlabeled data; as the name implies, it is a combination of supervised and unsupervised learning [10]. In a generic semi-supervised method, given a dataset of labeled and unlabeled data, instances are handled in one of two ways:

1. Labeled data points follow the same steps as in conventional supervised learning: predictions are generated,

losses are computed, and network weights are updated using gradient descent.

2. The model uses unlabeled data points to help it produce predictions that are more reliable and consistent. Unlabeled examples are utilized to advance the progress of labeled examples, either via pseudo-labels or by adding an additional unsupervised loss term.

FixMatch [11] utilizes both consistency regularization and pseudo-labels, while MixMatch [12] trains on both labeled and unlabeled data by combining mixup operations with label sharpening. Hashmi et al. [13] in the year 2020 published an article where he demonstrated deep transfer learning algorithms for automatic detection of pneumonia through chest X-ray images. The deep networks had complex structures with parameters and used less processing power to provide improved accuracy. Moreover, a weighted classifier was suggested as an effective way to integrate several topologies. The model's resilience was demonstrated by the various scores, including accuracy, precision, recall, and AUC, achieved during the tests. The suggested model demonstrates a high level of accuracy, with a precision of 98.857%. Its effectiveness was further validated by its exceptional AUC score of 99.809 and F1 score of 99.002 [14].

An automated computer-aided diagnosis (CAD) system is proposed for implementation in 2023 in an article that uses the "Chest X-Ray" dataset to assess the system's performance in diagnosing pneumonia [15]. The class imbalance problem can be solved by merging the DCGAN and WGAN-GP networks, which stand for Wasserstein GAN gradient penalty. The suggested approach creates a sample for the minority class, or "pneumonia." In contrast, the majority class, or "no-findings," was subjected to Random Under-Sampling (RUS). The transfer learning methodology is then applied to train three cutting-edge CNN models on the balanced data [16].

Kundu et al. [17] introduce an automated computer-aided design (CAD) system that utilizes deep learning for categorizing chest X-ray images as either "Normal" or "Pneumonia" to assist the medical professionals. A weighted average was formed by utilizing the decision scores of GoogLeNet, DenseNet-121, and ResNet-18 in an ensemble architecture [18]. A novel approach combined accuracy, recall, F1-score, and AUC to determine classifier weights by utilizing the hyperbolic tangent function. The system produced an accuracy rate of 98.81%, a precision rate of 98.82%, a sensitivity rate of 98.80%, and an F1-score of 98.79% by using a fivefold cross-validation strategy on the Kermany dataset. It achieved rates of 86.86%, 87.02%, 86.89%, and 86.95% on the RSNA challenge dataset, respectively.

Manickam et al. [19] performed pre-processing on chest X-ray images for detecting pneumonia. Author applied the U-Net architecture-based segmentation technique to identify

abnormal regions. The images were then identified as normal or abnormal (bacteria, viral) by utilizing pre-trained models from the ImageNet dataset, i.e., InceptionResNetV2, InceptionV3, and ResNet50. The Adam and Stochastic gradient descent (SGD) optimizers are employed and evaluated using batch sizes of 16 and 32. The pre-trained models, including RetinaNet + Mask RCNN, fully connected RCNN, VGG16, DenseNet-169 + SVM, and Xception, are compared to other CNN models based on their values using various measures [20]. The ResNet50 model accomplished a precision of 93.06%, recall of 96.78%, precision of 88.97%, and F1-score of 92.71% by surpassing the performance of the other models.

Mabrouk et al. [21] recommend three well-known convolutional neural networks (CNNs) for the classification of images: MobileNetV2, DenseNet169, and Vision Transformer. Fine-tuning trains these models using chest X-ray data. Throughout the experiment, these three models' retrieved features were combined to get the findings. The proposed EL strategy surpasses advanced methods with 93.91% accuracy and 93.88% F1-score during testing.

Zang et al. [22] applied the dynamic histogram equalization (DHE) technique in this research to improve the contrast of the images. Afterwards, they developed a simple convolutional neural network (CNN) model using the VGG architecture. This model is designed to extract features either from original photos or earlier feature maps. The model comprised six layers in total, which included the max-pooling layers, ReLU activation function, and drop operation. The achieved accuracy rate of 96.07% and precision rate of 94.41% indicate that the proposed model surpasses advanced CNN model designs.

Bhatt and Shah [23] explain the deep learning application for classifying digital chest X-ray images to identify the absence or presence of pneumonia. A composite framework extracts scores from three models with different kernel sizes and employs a weighted average method to provide a final numerical output. To classify the final sigmoid output, a threshold value was chosen. High recall of 99.23% was crucial for the issue, since the precision (80.04%) and accuracy (84.12%) were below domain standards more research needed to be done.

In 2020, Chandra and Verma [24] extracted the features of the restricted region surrounding the lungs, known as the region of interest (ROI), as these pixels had a greater impact on detecting pneumonia compared to the surrounding areas. The suggested technique has been tested with five well-established classifiers: multilayer perceptron, logistic regression, random forest, and classification by regression. The study utilizes 412 chest X-ray images from the ChestX-ray14 dataset, consisting of 206 images of normal patients and 206 images of patients with pneumonia. The suggested approach is compared to the prevailing method using standardized

classifiers. Experiments show that the suggested approach outperforms the prior method with 95.63% logistic regression classifier accuracy and 95.39% multilayer perceptron accuracy.

The study [25] examines a Kaggle dataset of chest X-rays to find algorithms that can diagnose pneumonia using the Vision Transformer (ViT) architecture. Using a hybrid of self-attention processes and transformer architecture, the ViT model is able to extract global context and spatial links from chest X-ray images. The results demonstrate that the framework based on Vision Transformer detects pneumonia from chest X-rays with a sensitivity of 95% and a specificity of 98%. The accuracy rate is 97.61%. When dealing with multi-resolution images, global context, and geographical linkages, the ViT model excels. Clinical pneumonia diagnosis systems based on convolutional neural networks (CNNs) are surpassed by this technology.

An X-ray image pre-processor, a threshold segmentation technique, the YOLOv3 detector, SVM, and SoftMax are all part of the deep learning model introduced by Asnake et al. [26]. The model then uses these features to distinguish between normal lung conditions and pneumonia. The proposed model was trained and tested by using chest X-ray images. A 99% F1-score, recall, precision, and accuracy level are indicated by all values. The results show that deep characteristics can reliably and successfully identify pneumonia cases. The gaps presented in each study are concisely summarized in Table 1, which emphasizes their contribution to the rationale for the proposed approach to enhance the robustness, generalizability, and demographic inclusivity of pneumonia detection models.

Methodology

Datasets

The datasets containing chest X-ray images related to pneumonia have become valuable resources for developing computer-aided diagnostic systems. In this study, we explore three different but related datasets, the Chest X-Ray Images (Pneumonia) dataset, a Brazilian labeled chest X-ray dataset (BRAX), and CheXpert, which provides a large number of chest X-ray images for pneumonia detection.

Chest X-Ray Images (Pneumonia)

The first dataset is the Chest X-Ray Images (Pneumonia) dataset taken from Kaggle [27]. It is consolidated into three folders: train (5216), test (624), and val (16). These folders have subfolders for each image category, namely pneumonia and normal. The collection contains a total of 5856 X-ray images from a hospital in Guangzhou, China, that have been

Table 1 Gaps/limitations of the study

Author/paper	Gaps/limitations of the study
Cancer Genomics Study [3]	Lack of diversity in datasets, primarily representing Caucasians, leading to potential biases in model generalization and healthcare disparities
Park et al. (2023) [4]	Limited focus on model adaptation to diverse demographics, which may impact generalizability in clinical settings
Labhane et al. [5]	High accuracy achieved, but no assessment of performance across varied demographics or exploration of techniques for improved generalization
Medical Imaging Disparity Study (2022) [7]	Noted sensitivity of models to demographic factors (age, sex, geography), but no specific methods proposed to leverage this for enhanced performance
Semi-Supervised Learning (SSL) Methods [11, 12]	Limited exploration of advanced SSL methods like FixMatch and MixMatch, which could better leverage unlabeled data in medical image analysis
Hashmi et al. (2020) [13]	Deep transfer learning demonstrated effectiveness but lacked consideration for class imbalance and dataset quality variability
Chest X-Ray8 Study (2023) [15]	Addressed class imbalance with GANs but did not evaluate effects on performance across diverse demographics, potentially limiting model robustness
Kundu et al. [17]	High accuracy achieved, but lacking in-depth evaluation of generalizability across patient demographics
Manickam et al. [19]	Did not analyze model performance with noisy or lower-quality images, which are common in real-world clinical scenarios
Mabrouk et al. [21]	Demonstrated effective fine-tuning but did not assess generalization across different patient demographics, which can impact model reliability
Zang et al. [22]	High accuracy achieved, but advanced techniques like SSL or demographic-specific transfer learning were not explored
Bhatt & Shah [23]	Achieved high recall but reported low precision and accuracy, signaling a need for more refined methods to optimize both sensitivity and specificity
Chandra & Verma [24]	Focused on small regions of interest but did not assess the robustness of classifiers across a larger, more diverse dataset
Vision Transformer Study [25]	Showed high capability but was not tested on multi-resolution or diverse, real-world datasets, limiting broader clinical applicability
Asnake et al. [26]	Optimized model architecture, but did not assess influence of demographic factors on accuracy and robustness, essential for fair and accurate healthcare outcomes

categorized as 4273 pneumonia and 1583 normal. The training folder consists of 5216 images, 26% of which are normal and 74% pneumonia. On the testing set, 62% of the chest X-rays are pneumonia and 38% normal. At the same time, the validation set contains 50% for both labels.

BRAX, a Brazilian Labeled Chest X-Ray Dataset

The second dataset is BRAX, a Brazilian labeled chest X-ray dataset [28]. It is comprised of 24,959 chest radiography studies obtained from a major general hospital in Brazil, consisting of a total of 40,967 images. All images have undergone verification by trained radiologists and were de-identified to ensure the privacy of the patients. The dataset features 14 different labels, including pneumonia, which were obtained through the use of Natural Language Processing methods employed to the free-text radiology reports written in Brazilian Portuguese. Figure 1 illustrates the distribution of the label in the BRAX dataset, which shows that most images have no finding, and only the 170 chest X-ray image was labeled as positive for pneumonia.

CheXpert

The third dataset we look at in this study is CheXpert [29], which is a substantial dataset of chest X-ray images and their correlated radiology descriptions devised to accelerate the enhancement of machine-learning models to discover and establish diverse pathologies in the chest, including pneumonia [26]. The dataset consists of 224,316 chest radiographs of 65,240 patients from Stanford Hospital and includes 14 different pathologies similar to BRAX as shown in Fig. 2. Each report was manually annotated by radiologists and labeled for uncertainty and a senior radiologist adjudicated disagreements between radiologists.

Data Preprocessing

Reading image data is a critical step in developing any image-processing application. Python provides several libraries to read and manipulate image data, including OpenCV, Pillow, and scikit-image. OpenCV is a popular and powerful library that provides various image processing

Fig. 1 Label Distribution in BRAX dataset

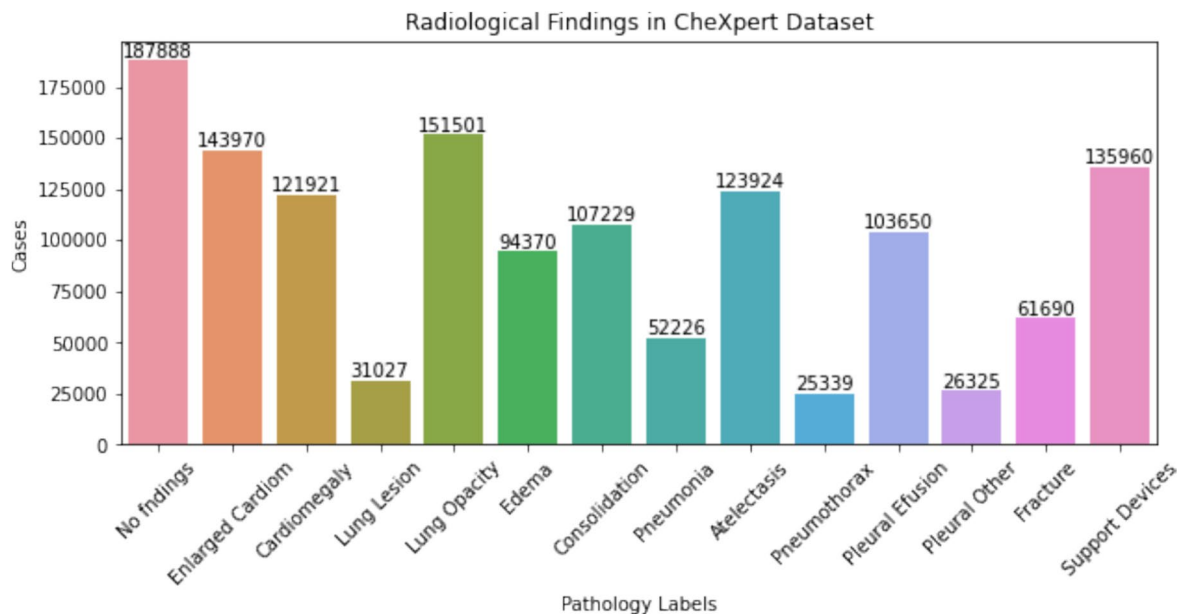
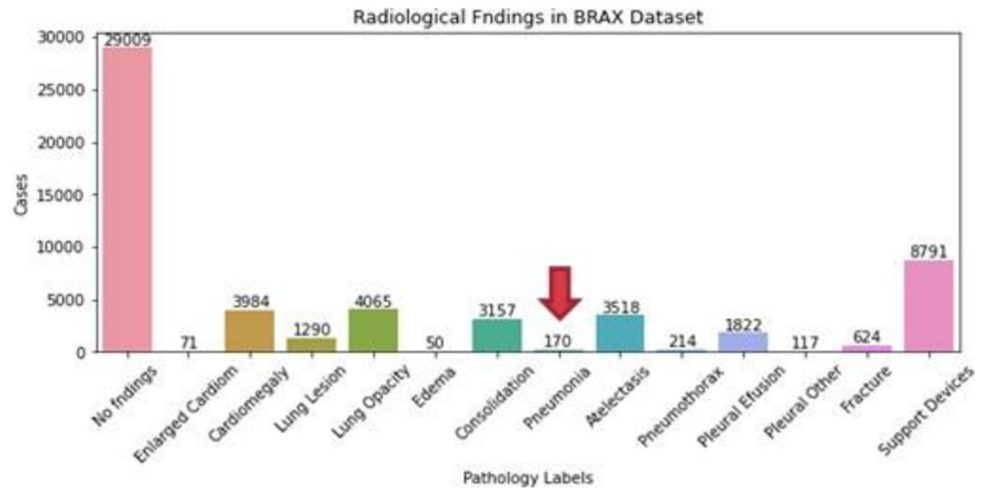


Fig. 2 Label distribution in CheXpert dataset

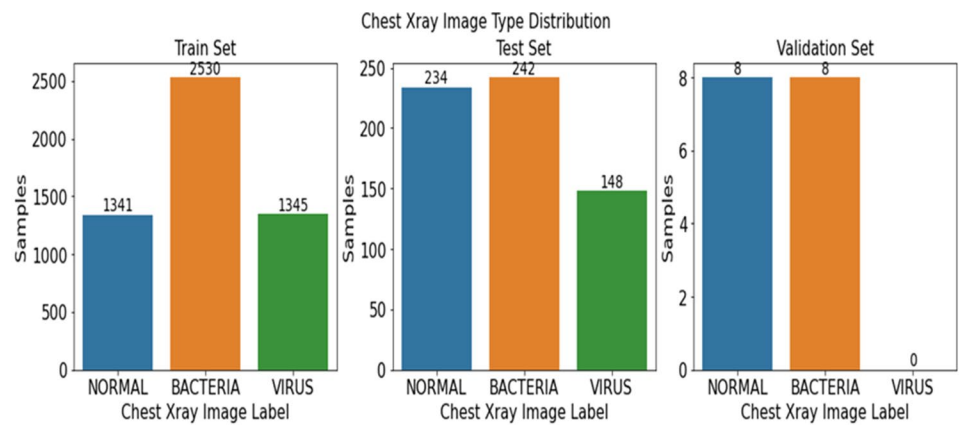
functions and can handle multiple image formats such as JPEG, PNG, TIFF, and BMP. The pillow is a fork of Python Imaging Library (PIL) and supports fewer image formats than OpenCV but provides a simple and easy-to-use interface for basic image processing tasks. Scikit-image is another popular library that provides a rich set of image-processing functions for scientific computing. It can handle various image formats and provides filtering, segmentation, feature extraction functions, and more. To read image data in Python, first, we use the OpenCV (cv2) and os packages to traverse directories and read images from specific folders. Then, we use the cv2.imread() function to read each image from a specified file path. Once the images are loaded, we store them in Data Frames for processing. To enhance

computational efficiency and facilitate training, high-resolution images of diverse sizes were resized to a uniform 224 by 224 pixels.

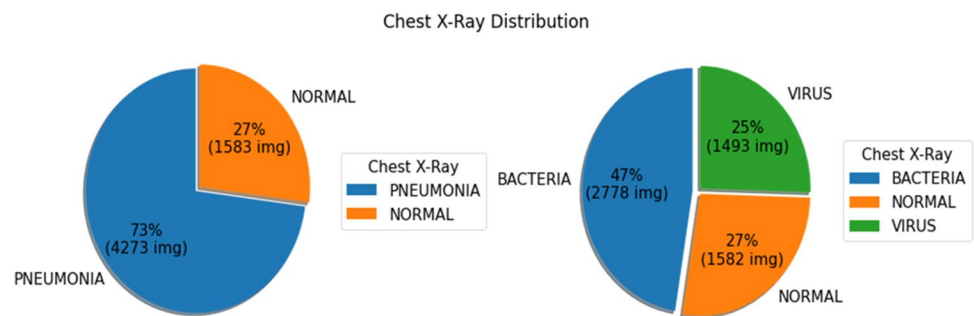
Chest X-Ray Images (Pneumonia) EDA

During reading the Chest X-Ray Images (Pneumonia) dataset, we noticed that image captions referred to normal, bacteria, and virus. Therefore, we create two different label attributes to treat this dataset as binary classification pneumonia (bacteria and virus) or normal. On the other hand, we can handle it as multiclass classification bacteria or virus or normal. Figure 3a and b shows that this dataset is imbalanced, which will result in highly biased results.

Fig. 3 Data distribution in Chest X-Ray Images (Pneumonia) dataset. **(a)** Dataset Separated Folders (Multiclass). **(b)** Combined Folders



(a) Dataset Separated Folders (Multiclass)



(b) Combined Folder

Additionally, we visualize some samples of each class, including binary classes, as shown in Fig. 4a and multiclass as in Fig. 4b.

BRAX, a Brazilian Labeled chest X-Ray Dataset EDA

The BRAX dataset assigned positive (0.1) and negative (0.0) for 13 diseases, including pneumonia. However, there

were only 170 images labeled as positive for pneumonia. In addition, we found that some images were defective and not perfect for training the models. Therefore, we excluded these images and ended up with 148 chest X-rays for pneumonia. This finding led us to treat this dataset as a balanced dataset by considering 148 images from both classes, pneumonia and normal. Then, we split the dataset into three sets: training (103, 103), validation (27, 27), and testing (18, 18), as shown in Fig. 5.

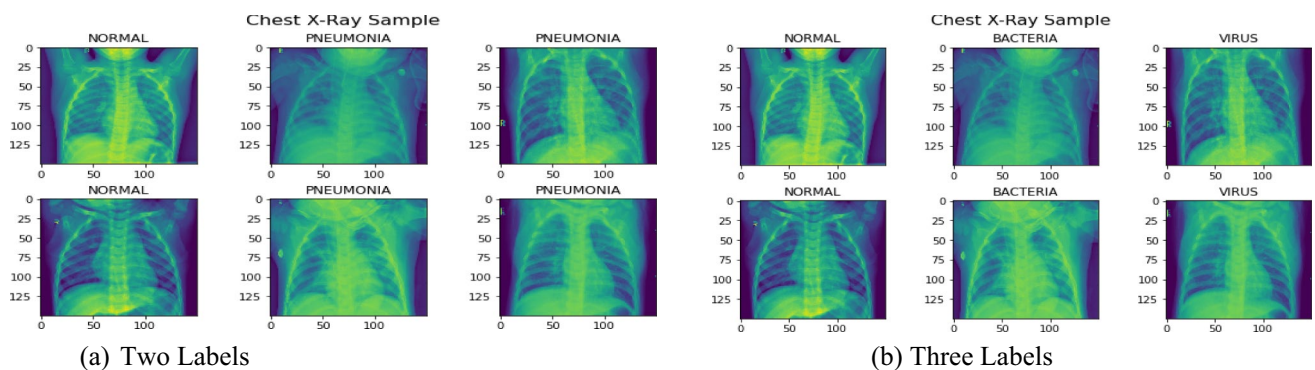


Fig. 4 Samples of chest X-ray images in each class. **(a)** Two Labels. **(b)** Three Labels

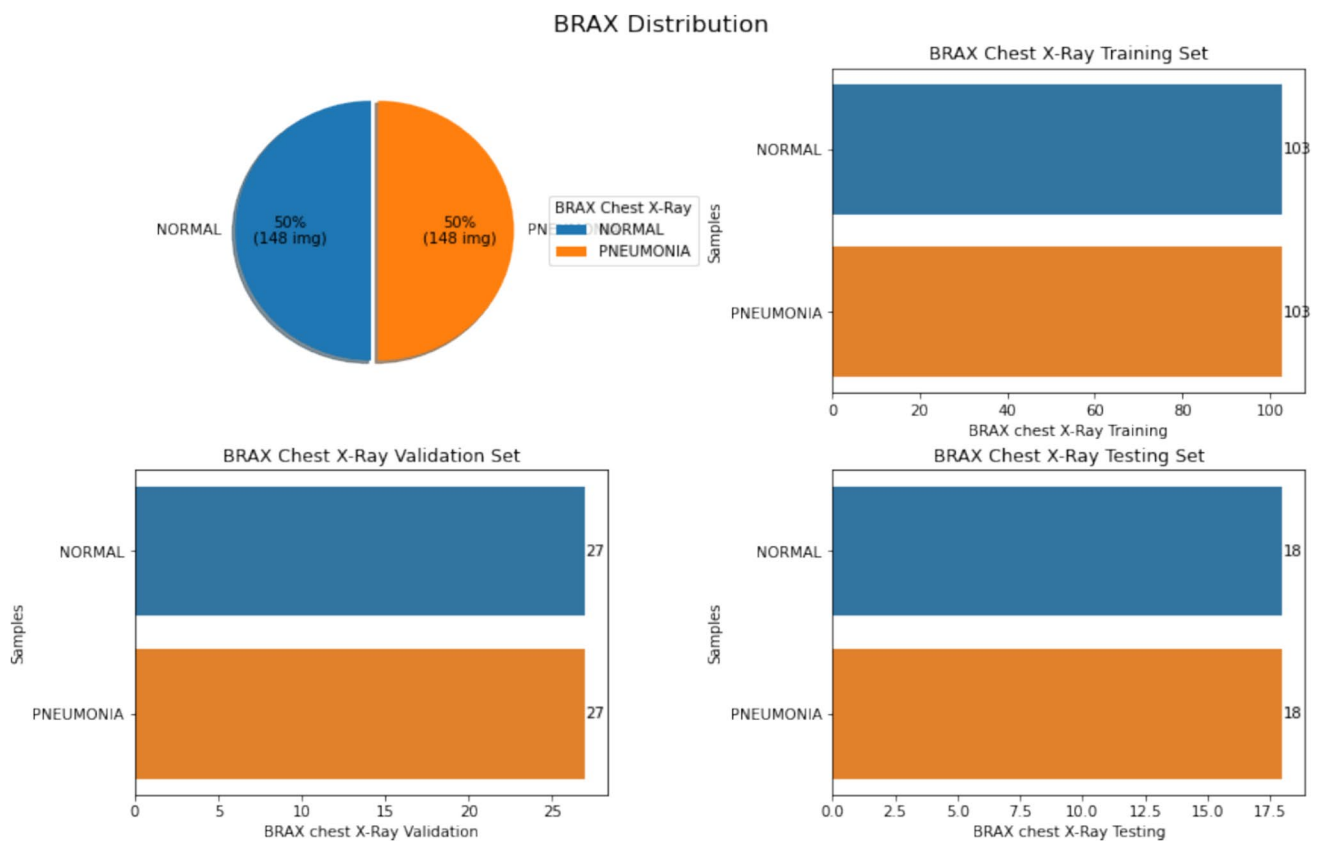


Fig. 5 Pneumonia distribution in BRAX dataset

CheXpert EDA

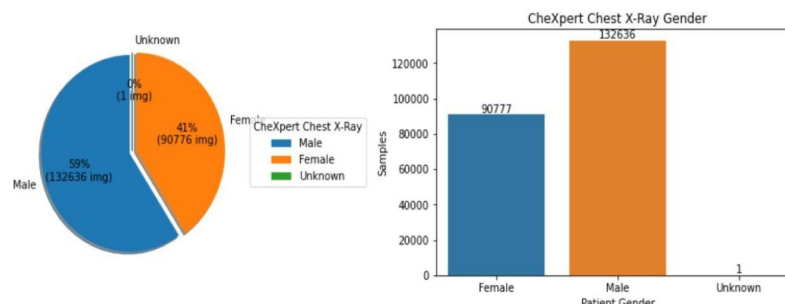
CheXpert dataset includes uncertainty labels for each disease label, indicating when radiologists had difficulty making a definitive diagnosis. The dataset was designed to help address the challenge of imbalanced classes and label uncertainty in medical image classification and to facilitate the development of machine learning algorithms that can accurately detect and diagnose thoracic diseases. Additionally, the CheXpert dataset contains statistics about patients' genders and ages. Chest X-ray contains 132,636 (59%) images from males and 90,776 (41%) from females that range from 18 to 107 years old. Figure 6a in the top left demonstrates that 77% of the images were normal and 23% were positive for pneumonia. Figure 6b in the top right shows that 30,841 (59%) of males in this experiment experience pneumonia disease and 21,383 (41%) of females experience the same disease. The pneumonia cases increased between the age of 58 and 68 as shown in Fig. 6c.

Imbalanced Dataset

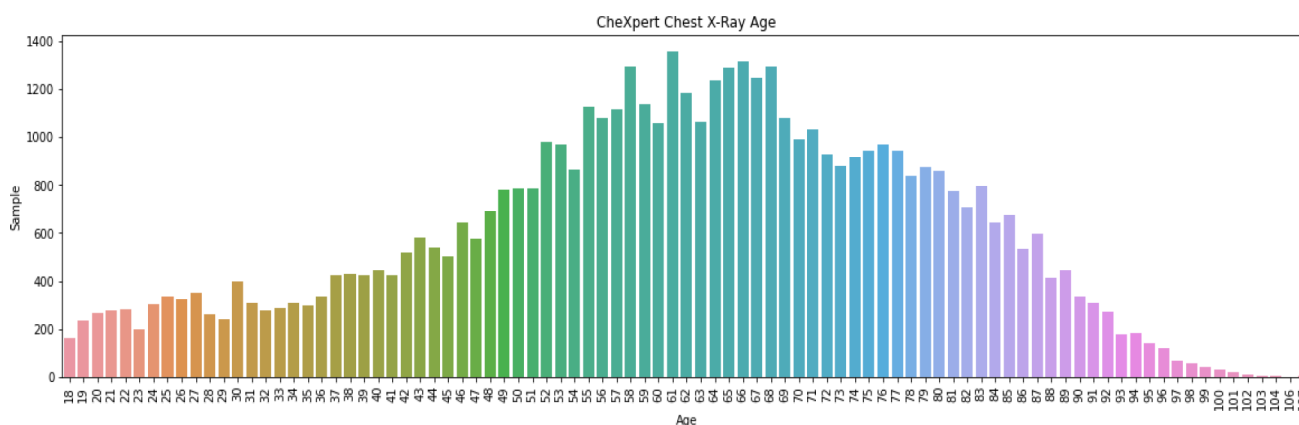
An absence of necessarily enormous, distinct collections of training data usually ends in models that overfit on the

training data and simplify inadequately. Classifying medical imaging data is equally costly and slow; an absence of necessarily varied categorized training facts is amongst the key obstacles to getting models implemented for clinical service. Additionally, medical data repeatedly experiences the difficulty of classification unevenness. We are also dealing with such a problem in the Chest X-Ray Images (Pneumonia) dataset. To resolve this problem, there are several ways. One method is to apply various changes to the original dataset to generate more data from the minority class. Along with the addition of Gaussian noise, these transformations also include random flips, rotations, and translations. These techniques exploit changes that are known not to affect the class, but they only provide a small amount of realistic alternative data. A better solution to address the unbalanced dataset involves using generative models, such as variational autoencoders (VAEs), generative adversarial networks (GANs), and diffusion models, which can approximate the data distribution. However, these models are computationally intensive and require high-performance computers with powerful GPUs. As an alternative, this study elaborates on the use of balanced weights and combining them with data augmentation techniques for addressing imbalanced datasets.

(a) Pneumonia Label Distribution



(b) Gender Distribution



(c) Age Distribution

Fig. 6 CheXpert dataset statistics. **(a)** Pneumonia Label Distribution. **(b)** Gender Distribution. **(c)** Age Distribution

Models and Dataset Split

In this project, we investigate the performance of three of the most popular image recognition models, including visual geometry group (VGG), deep residual networks (Resnet), and Vision Transformers (ViT). Specifically, we implemented the VGG16 model that consists of 13 convolutional layers and three fully connected layers which takes in an image input size of 224×224 . There are numerous ResNet variants that operate on the same concept but have varying numbers of layers. We examine the Resnet50 model that denotes the variant capable of functioning with 50 neural network layers. Moreover, the Vision Transformer (ViT) model was implemented. It is a visual paradigm based on the architecture of a text-based task-specific transformer. The

ViT model represents an input image as a series of image patches, similar to the series of word embeddings used when applying transformers to text, and predicts image class labels directly [30].

After data preprocessing, we split the datasets into three sets for training, validating, and testing. Table 2 represents the distribution of each class in each set for the three datasets: Chest X-Ray Images (Pneumonia), BRAX, a Brazilian labeled chest X-ray dataset, and CheXpert dataset [31].

Mixture Learning and Independent Learning

Mixture learning and independent learning are standard methods of training models off of biomedical datasets, so we wanted to run those experiments alongside our other

Table 2 Datasets split for train, validate, and test

	Dataset	Kaggle	BRAX	CHEX
Train	Normal	1320	103	12182
	Pneumonia	2509	103	2092
Validate	Normal	29	27	696
	Pneumonia	29	27	696
Test	Normal	233	18	696
	Pneumonia	389	18	696

experiments to compare them. The mixture learning model would be created by collating all three datasets to train the model off. The expected results here would be the smallest dataset suffering the most in terms of accuracy, but we would still want to obtain the results for each model we are testing to have a default accuracy to work with. The independent learning serves a similar purpose, being individual runs of a model on a singular dataset [32].

Semi-supervised Method

To leverage the large amount of unlabeled data available in the CHEX dataset for pneumonia detection, we have decided to employ semi-supervised learning techniques. The dataset contains approximately 16,000 unlabeled instances, which have not been conclusively identified as either positive or negative for pneumonia due to detection uncertainty [33].

1. We employed a simple strategy for performing semi-supervised learning:
2. We train the neural network on the labeled dataset for a few epochs to achieve a high accuracy on it.

3. Next, we determine the pseudo-labels for the unlabeled dataset.

4. We incorporate the unlabeled data with a probability of over 90% into the training data.

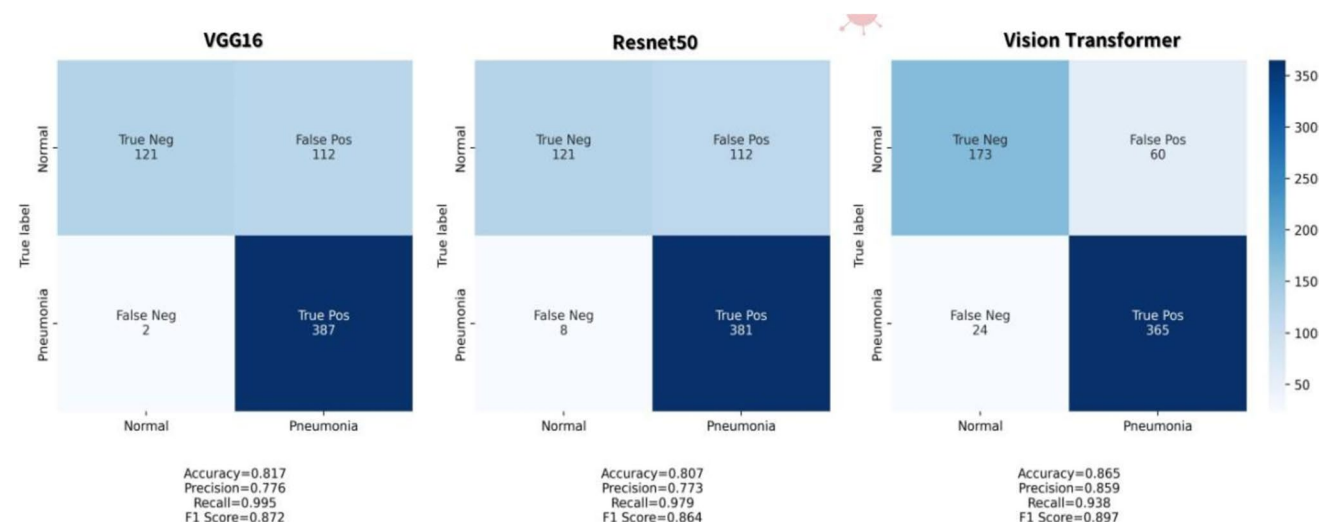
We return to step 1 and repeat this process until the validation loss stops improving. We apply this algorithm on Kaggle and CHEX datasets.

Results

In this section, we will begin by comparing the performance of various deep learning models for pneumonia detection. Next, we will implement different strategies to address the challenges of imbalanced datasets. Following that, we will evaluate the zero-shot and few-shot learning experiments across our datasets. Finally, we will explore the application of semi-supervised learning to leverage the unlabeled data.

Model Performance Evaluations

We investigated the performance of VGG16, Resnet50, and ViT and evaluated their accuracies on three different datasets: Kaggle, BRAX, and CheXpert. We trained three models on the Kaggle dataset, which is a large dataset, and applied early stopping on the validation dataset to prevent overfitting. The initial results are presented in Fig. 7. We observe that the Vision Transformer outperforms other methods in terms of accuracy. However, when considering recall, VGG16 demonstrates superior performance compared to the other methods.

**Fig. 7** Initial results for the Kaggle dataset

Mitigating Imbalance in Datasets

In this section, we present the results of our algorithms in addressing the challenges posed by imbalanced datasets, a common issue in pneumonia detection.

Kaggle Dataset Without Class Weights or Data Augmentation

We will optimize the models from the previous section by incorporating class weights and applying data augmentation to enhance accuracy.

The statistical analysis of the Kaggle dataset reveals that the comparison between VGG16 and Vision Transformer models shows a substantial difference, with a statistic t of 15.1221 and a p -value of 0.0001 ($p < 0.05$). Similarly, Vision Transformer and ResNet50 exhibit a significant distinction with the same statistical results ($t = 15.1221$, $p = 0.0001$). Conversely, the comparison of VGG16 and ResNet50 models yields a statistic t of 0.0045 with a p -value of 0.947, indicating no statistically significant difference between them ($p > 0.05$). These findings suggest that, within this dataset, Vision Transformer and ResNet50 outperform or differ significantly from VGG16 when class weights or data augmentation are not introduced.

Class Weights

Class weights are a relatively simple way to deal with imbalance in classes in datasets. For validation and testing datasets, we generally take an even sample of both classes to avoid any confounding variables there, but this is not the case for the training datasets. We have the option there to do a similar sampling and exclude some of the data

we have for the training dataset, or we can take all of the leftover data and train using that. The issue there would be that with the larger datasets, such as the Kaggle and the CHEX datasets, the imbalance in the classes would cause issues with the classification of the minority class because of just how much more of the majority class there would be present since we are working on the scale of thousands of images at once. Class weights simply assign a heavier weight to the minority class in the loss function, punishing mistakes in predictions on the minority class more heavily than it would the majority class, leading to a more accurate model for the minority class than would otherwise form. For the Kaggle dataset, a 4:1 ratio was seen between the normal and pneumonia images, and so a weight of 4 was placed on the pneumonia images. For the CHEX dataset, the pneumonia positive class was once again the minority class with the images showing a 6:1 ratio this time, and the classes were appropriately weighted once again. The effect of the class weights can be seen with the Kaggle dataset in Fig. 8. It is evident that the performance of all methods has improved.

The statistical analysis of model comparisons on the Kaggle dataset with balanced weight presents varying significance levels. The comparison between VGG16 and Vision Transformer models, with a statistic t of 7.0866 and a p -value of 0.0078, indicates a significant difference ($p < 0.05$). Furthermore, the comparison between Vision Transformer and ResNet50 reveals a highly significant difference, with a statistic t of 21.0974 and a p -value of $4.37e-06$ ($p < 0.001$). However, the comparison between VGG16 and ResNet50 models shows a statistic t of 3.6541 with a p -value of 0.056, indicating no statistically significant difference ($p > 0.05$). These results suggest that Vision Transformer differs significantly from both VGG16

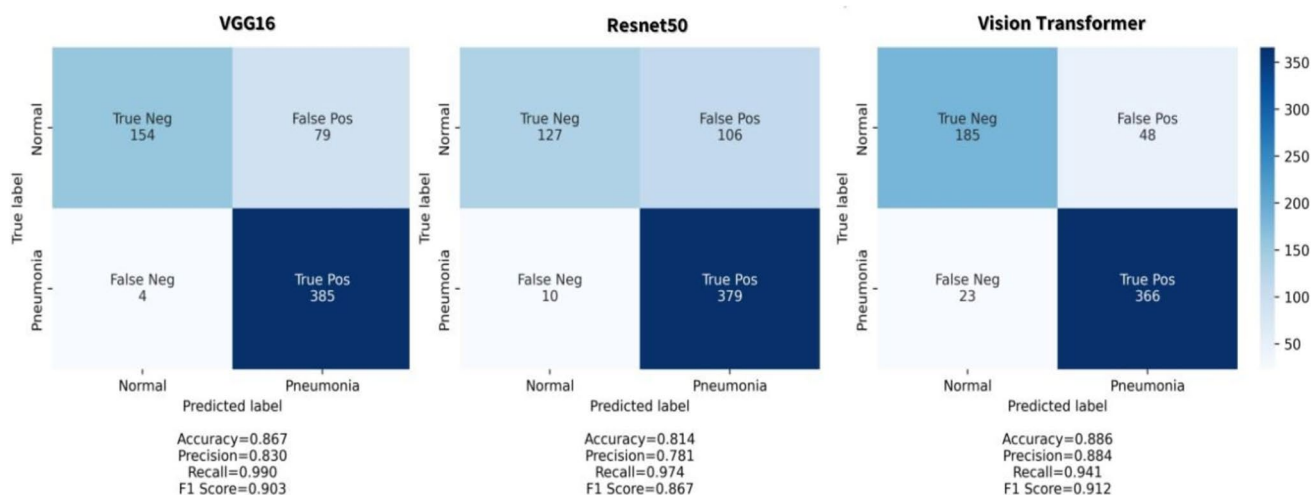


Fig. 8 Results with balanced weights

and ResNet50, highlighting its distinct performance characteristics on the Kaggle dataset with balanced weight.

Data Augmentation

Data augmentation is a popular technique used to address the issue of imbalanced datasets in machine learning. By generating more data from the minority class, data augmentation can help to improve the performance of machine learning models that are trained on imbalanced datasets. Common transformations used in data augmentation to create additional variations of the minority class include random translations, rotations, and flips, as well as the addition of Gaussian noise. While data augmentation can be effective in improving the performance of machine learning models on imbalanced datasets, it is important to be cautious about overfitting and to validate the results carefully.

The parameters chosen for data augmentation are as follows (Table 3):

By employing data augmentation, we created 1000 additional instances for normal cases and 300 augmented images for pneumonia cases. We evaluated the performance of the resulting model on the test dataset and listed out the results in Fig. 9.

Table 3 Parameters for data augmentation

Parameter	Value
Rotation range	30
Width shift range	0.1
Height shift range	0.1
Shear range	0.2
Zoom range	0.2

We decided to apply data augmentation to the majority class in addition to the minority class due to poor performance when only augmenting the minority class. We hypothesized that this could be due to the model's lack of exposure to transformations and noise in the majority class, leading to confusion.

From the outcomes in Fig. 9, it is obvious that utilizing data augmentation along with balanced weight can lead to better performance on the test dataset compared to previous section.

The analysis of model performance on the Kaggle dataset with data augmentation reveals both significant and non-significant differences across model comparisons. The VGG16 and ResNet50 models exhibit a statistically significant difference, with a test statistic t of 9.7143 and a p -value of 0.0018 ($p < 0.05$). However, the comparison between VGG16 and Vision Transformer yields a statistic t of 0.0494 and a p -value of 0.8241, indicating no significant difference ($p > 0.05$). In contrast, Vision Transformer and ResNet50 demonstrate a significant difference with a statistic t of 11.8017 and a p -value of 0.0006 ($p < 0.05$). These findings suggest that ResNet50 performs distinctly from both VGG16 and Vision Transformer, while VGG16 and Vision Transformer perform comparably on this dataset when applying data augmentation.

Comparison of Kaggle Dataset Results

The present study assesses the execution of several deep learning models on an imbalanced dataset, employing various approaches incorporating class weights and data augmentation. The results are presented in Table 4, where it is observed that the Vision Transformer (ViT) model outperformed VGG16 and Resnet50 with an accuracy of 86.7%



Fig. 9 Results with data augmentation and balanced weights

Table 4 Results for different strategy applied on imbalanced Kaggle dataset

			Accuracy	Precision	Recall	F1-Score
	Regular	VGG16	0.817	0.776	0.995	0.872
		Resnet50	0.807	0.773	0.979	0.864
		Visual Transformer	0.865	0.859	0.938	0.897
	balanced weights	VGG16	0.867	0.830	0.990	0.903
		Resnet50	0.814	0.781	0.974	0.867
		Visual Transformer	0.886	0.884	0.941	0.912
	Augmentation + balanced weights	VGG16	0.910	0.899	0.964	0.931
		Resnet50	0.844	0.827	0.949	0.884
		Visual Transformer	0.891	0.902	0.925	0.914

when no techniques were employed to address the class imbalance. However, VGG16 exhibited the best accuracy of 91% when class weights and data augmentation techniques were combined. These findings suggest that the implementation of suitable techniques to handle class imbalance can enhance the implementation of deep learning models on imbalanced datasets.

The statistical analysis on the Kaggle dataset reveals significant differences among various configurations of the VGG16, ResNet50, and Vision Transformer models, with some configurations showing meaningful performance distinctions. Among VGG16 configurations, VGG16 without class weights or data augmentation and VGG16 with balanced weights differ significantly, with a statistic t of 5.3613 and a p -value of 0.0206 ($p < 0.05$), while VGG16 without class weights or data augmentation and VGG16 with data augmentation show a highly significant difference (statistic $t = 30.9156$, $p < 0.001$), as do VGG16 with balanced weights and VGG16 with data augmentation (statistic $t = 10.7107$, $p < 0.05$). For ResNet50, ResNet50 without class weights or data augmentation and ResNet50 with balanced weights exhibit no significant difference (statistic $t = 0.1147$, $p > 0.05$), but ResNet50 without class weights or data augmentation and ResNet50 with data augmentation (statistic $t = 6.1164$, $p < 0.05$) and ResNet50 with balanced weights and ResNet50 with data augmentation (statistic $t = 4.2842$, $p < 0.05$) differ significantly, highlighting distinct performance in ResNet50 with data augmentation compared to the other configurations. In the Vision Transformer model, Vision Transformer without class weights or data augmentation and Vision Transformer with balanced weights (statistic $t = 1.1204$, $p > 0.05$) and Vision Transformer with balanced weights and Vision Transformer with data augmentation (statistic $t = 0.7356$, $p > 0.05$) do not show significant differences, but Vision Transformer without class weights or data augmentation and Vision Transformer with data augmentation do (statistic $t = 4.0404$, $p < 0.05$). These findings suggest unique performance characteristics across model configurations, warranting further investigation into specific attributes driving these differences.

BRAX Dataset

We trained three models on the BRAX dataset, which is a small dataset, and applied early stopping on the validation dataset to prevent overfitting. Based on the results from the Kaggle dataset, we applied data augmentation and balanced class weights to address the dataset's imbalanced nature. The results, presented in Fig. 10, are compelling and demonstrate the effectiveness of our methodology to deal with an imbalanced dataset such as BRAX dataset.

The statistical analysis of the entire BRAX dataset without employing any balanced technique indicates no significant differences among the VGG16, ResNet50, and Vision Transformer models. The comparison between VGG16 and ResNet50 yields a statistic t of 1.125 with a p -value of 0.289 ($p > 0.05$), showing no statistically significant difference. Similarly, VGG16 and Vision Transformer show no significant difference, with a statistic t of 0.5714 and a p -value of 0.450 ($p > 0.05$). Additionally, the Vision Transformer and ResNet50 models are comparable, as indicated by a statistic of 0.0 and a p -value of 1.0. These results suggest that the models perform similarly on the BRAX dataset, indicating limited variability in their performance in this dataset without considering a balanced technique.

Mixture Learning

We trained all three models on a collated dataset composed of all three datasets with early stopping incorporated into the architecture. The results in Fig. 11 showcase just how badly the accuracy drops for the smallest dataset when mixture learning is done. An interesting, but logical, result of this was that the largest dataset benefited from the mixture learning method with an increase in its accuracy compared to any of its independent runs.

The statistical analysis of the mixture learning indicates significant differences in mixture learning between certain datasets. The CHEX and BRAX datasets show a highly significant difference, with a statistic t of 87.4589 and a p -value of $8.60e - 21$ ($p < 0.001$). Similarly, the Kaggle and BRAX

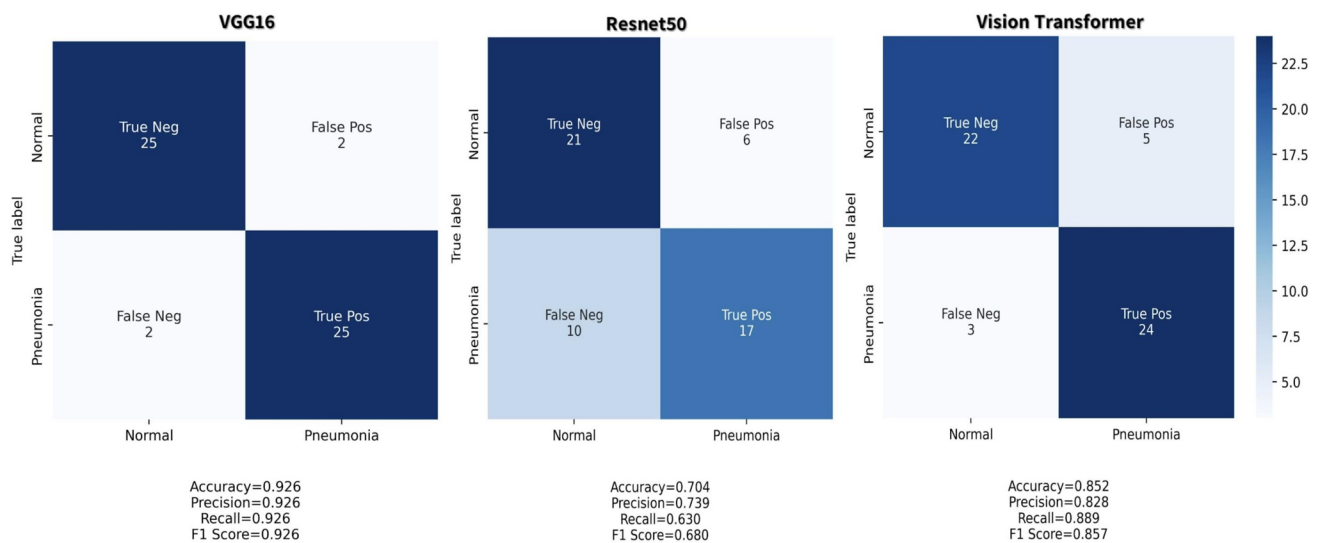


Fig. 10 Performance evaluation of models trained on the entire BRAX dataset

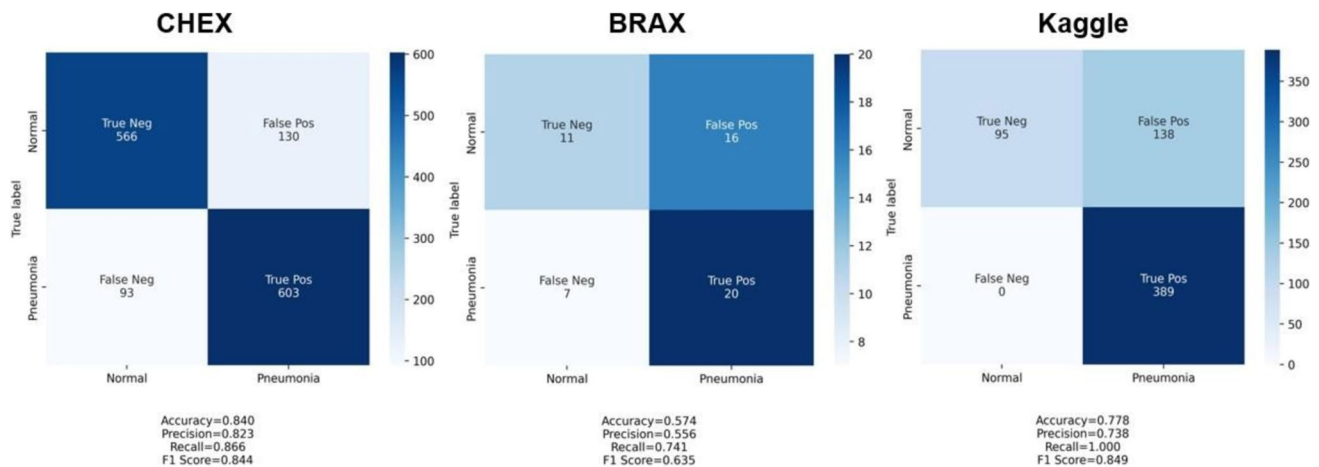


Fig. 11 Results on mixture learning on all three datasets

datasets differ significantly, with a statistic t of 95.0714 and a p -value of $1.84e-22$ ($p < 0.001$). In contrast, the CHEX and Kaggle datasets exhibit no significant difference in mixture learning, as indicated by a statistic t of 0.1828 and a p -value of 0.669 ($p > 0.05$). These results highlight considerable performance distinctions involving the BRAX dataset, suggesting unique characteristics in how it interacts within mixture learning compared to CHEX and Kaggle.

Transfer Learning from Kaggle Dataset to BRAX Dataset

As stated in our proposal, our target is to mark a significant aid in the discipline of biomedical image evaluation by exploiting transfer learning on X-ray image datasets obtained

from different geographical regions. Our success in creating an effective model for the Kaggle dataset, utilizing VGG16 architecture in combination with data augmentation and balanced weighting, has provided us with a solid foundation to begin implementing transfer learning techniques. To do so, we are going to use two types of transfer learning.

Zero-Shot Learning

Zero-shot learning is a type of transfer learning that involves training a model on a source task (in this case, the Kaggle dataset), without any prior data available for the target task (the BRAX dataset), and immediately testing its performance on the target task. This method is exceptionally handy

when coping with partial or diverse data, as it facilitates models to absorb and generalize among distinct domains.

The results depicted in Fig. 12 indicate that transfer learning is not proving to be effective in this context, as the model is not achieving a satisfactory level of accuracy.

Few-Shot Learning

Few-shot learning is a transfer learning practice that grants us to direct a model on a modest volume of data from the target task accompanied by a substantial amount of data from a shared resource task (Kaggle dataset). This methodology is exceptionally effective in circumstances where there is restricted data accessibility or the domain is extremely dedicated, as it lets the model influence previous facts and simplify to additional, invisible patterns. To employ few-shot learning, we first prepare the model on the source task and then modify it on some samples from the target task.

The results depicted in Fig. 13 demonstrate that as the size of the target dataset (BRAX) increases for fine-tuning, the accuracy of the model also improves. Interestingly, even with as few as 64 images from the BRAX dataset, we were able to achieve relatively acceptable levels of accuracy.

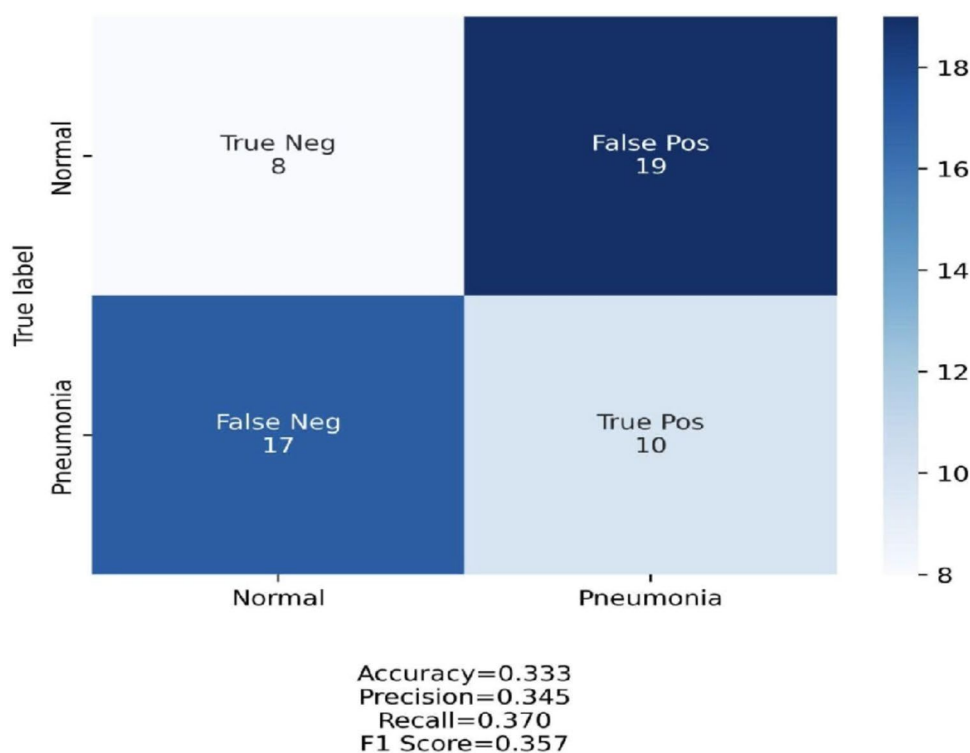
The statistical analysis on the few-shot learning dataset reveals no significant differences among various image configurations. Comparisons between N 16 (images) and N 32 (images), N 16 and N 64 (images), and N 64 and N 32 (images), all yield a statistic t of 0.25 with a consistent p -value of 0.6171 ($p > 0.05$). However, the comparison

between the zero-shot and few-shot indicates a significant difference between the zero-shot and few-shot models. With a statistic t of 12.1905 and a p -value of 0.0005 ($p < 0.001$), the results suggest that the two models exhibit distinct performance characteristics in this context. This finding highlights the effectiveness of the few-shot model over the zero-shot model, warranting further investigation into the underlying factors contributing to this difference in performance.

CHEX Dataset

With the results we obtained from testing on the Kaggle dataset, we started off with a simple application of the three models we had been testing on the CHEX dataset. The goal here would be to do independent learning with the CHEX dataset and obtain the best model possible here. The follow-up to this would be a transfer learning similar to what we did between the Kaggle and the BRAX datasets. For now, we only roped in the class weights consideration that we tested with the Kaggle dataset, but in the future, we will be incorporating data augmentation and possible artificially generated images via other methods as well. We see the results of this testing in Fig. 14. The validation and testing sets of the dataset were kept perfectly even between the two classes of pneumonia positive and pneumonia negative, but the training data caught all the leftover data which led to an imbalance of the two classes. To prevent one class from overpowering the other in terms of data, class weights of

Fig. 12 Results on BRAX dataset with zero-shot learning from Kaggle dataset



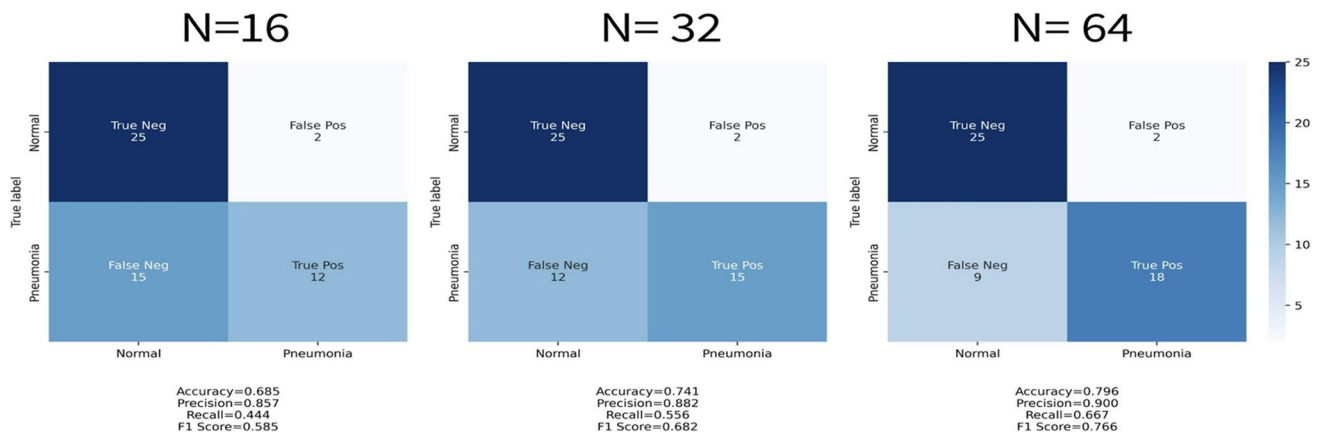


Fig. 13 Results on BRAX dataset with few-shot learning from Kaggle dataset; N is the number of images from BRAX dataset has been used to fine-tune the model

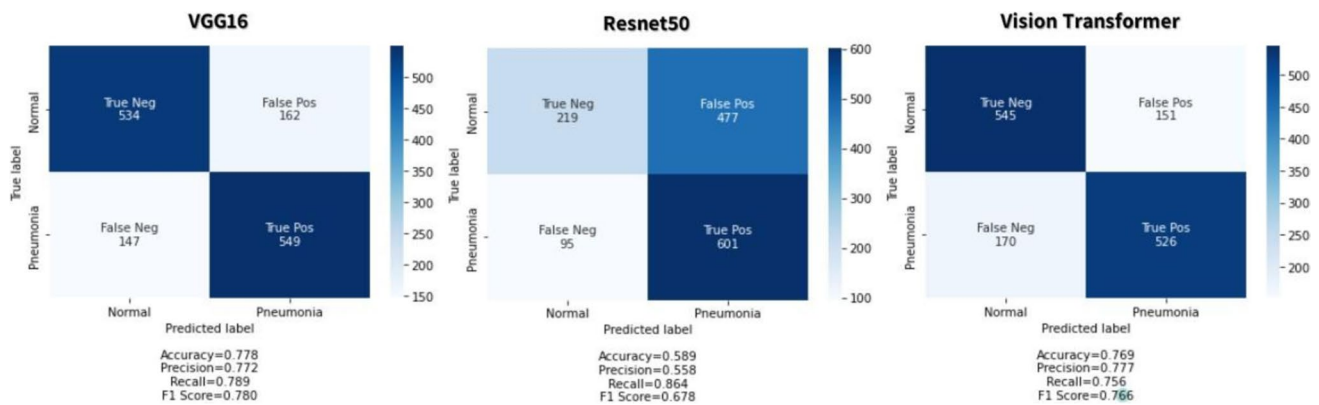


Fig. 14 Performance evaluation of models trained on the entire CHEX dataset

6:1 were used in accordance with the ratio of class 1:class 2. The accuracy with the CHEX data here was overall lower than we would have liked, especially with the Resnet model, and the recall was in a similar place. Further fine tuning would be required on the data to get better results such as the incorporation of data augmentation.

Semi-supervised Learning

To examine the effect of semi-supervised learning on classification performance, we utilized two methods and compared their improvements. The Mean Teacher algorithm was used to guide a VGG16 model on mutually labeled and unlabeled data with the Kaggle dataset. The model was trained for 50 epochs with a batch size of 32 and a learning rate of 0.001. Early stopping was incorporated here as well as class weights, and the VGG model was utilized since that saw the best results in previous experiments. A much simpler method of self-training incorporates a limited number of high-confidence unlabeled data (90% confidence or

above) after every epoch on the CHEX dataset. This method utilized the same hyperparameters as the previous method as well as the same VGG model to remove any unwanted variables affecting the comparison. Aside from the internal comparisons, we also wanted to compare the methods to the results we saw with supervised learning to see if there was significant increase in accuracy meriting further research into the matter.

The analysis of the semi-supervised learning on the CHEX dataset reveals significant performance differences among the VGG16, ResNet50, and Vision Transformer models. The comparison between VGG16 and ResNet50 shows a substantial difference, with a statistic t of 154.2973 and a p -value of $1.99e-35$ ($p < 0.001$), indicating that these models perform distinctly. In contrast, the VGG16 and Vision Transformer models do not show a significant difference, with a statistic t of 0.3195 and a p -value of 0.572 ($p > 0.05$). However, the Vision Transformer and ResNet50 models exhibit a significant difference, as evidenced by a statistic t of 168.1927 and a p -value of $1.84e-38$ ($p < 0.001$). These

results confirm the unique performance characteristics of the VGG16 and ResNet50 models compared to the Vision Transformer within the context of semi-supervised learning.

Kaggle Dataset

The Kaggle dataset did not have any unknown labels or uncertainties to work with, so we created a set of unlabelled data by using an initial 1000 labeled data points and consider the rest as a faux-unlabeled dataset. As Fig. 15 shows, the semi-supervised learning works much better than supervised learning here with a 10.9% increase in accuracy between the two methods.

The statistical analysis of the Kaggle dataset comparing supervised and semi-supervised models reveals a significant performance difference. With a statistic t of 39.2925 and a p -value of $3.65e - 10$ ($p < 0.001$), these results indicate that the supervised and semi-supervised models exhibit distinct characteristics in their performance. This finding emphasizes the effectiveness of one approach over the other, suggesting that the choice between supervised and semi-supervised learning strategies can have a considerable impact on model outcomes in this context.

CHEX Dataset

The CHEX dataset had images that were labeled as uncertain for pneumonia, so we incorporated all of them as part of the unlabeled dataset. With around 16,000 unlabeled data points, we applied semi-supervised learning and compared it with supervised learning. As the results show in Fig. 16, there was an increase in the accuracy by 7.4%.

Discussion

Highlights of contribution

- Semi-supervised learning for medical imaging: This study improves performance for pneumonia identification in limited labeled data, confirming the efficiency of semi-supervised learning for medical imaging. Labeled data is expensive and limited in medical imaging, especially for complex disorders; hence, these methods are innovative. The accuracy gains (10.9% for Kaggle, 7.4% for CHEX) show semi-supervised learning could solve this barrier.
- Domain-specific transfer learning: This study examined the impact of transfer learning on pneumonia diagnosis in smaller datasets, despite its widespread use in general-purpose datasets like ImageNet. This helps determine which model architectures and tactics work for domain transfer in medical settings, filling a gap in previous studies that missed feature transferability across domains and dataset sizes.

Gaps and Limitations of Prior Studies

- Realistic semi-supervised learning: This study utilized advanced semi-supervised techniques and real-world unlabeled data with uncertain labels (from CheXpert), unlike previous studies that relied on fully labeled datasets or pseudo-labeling approaches. This more accurately depicts clinical circumstances with ambiguous labels and unlabelled cases, paving the way for medical label uncertainty research.
- Impact of high-confidence pseudo-labeling: Setting a 90% confidence threshold for pseudo-labels in self-train-

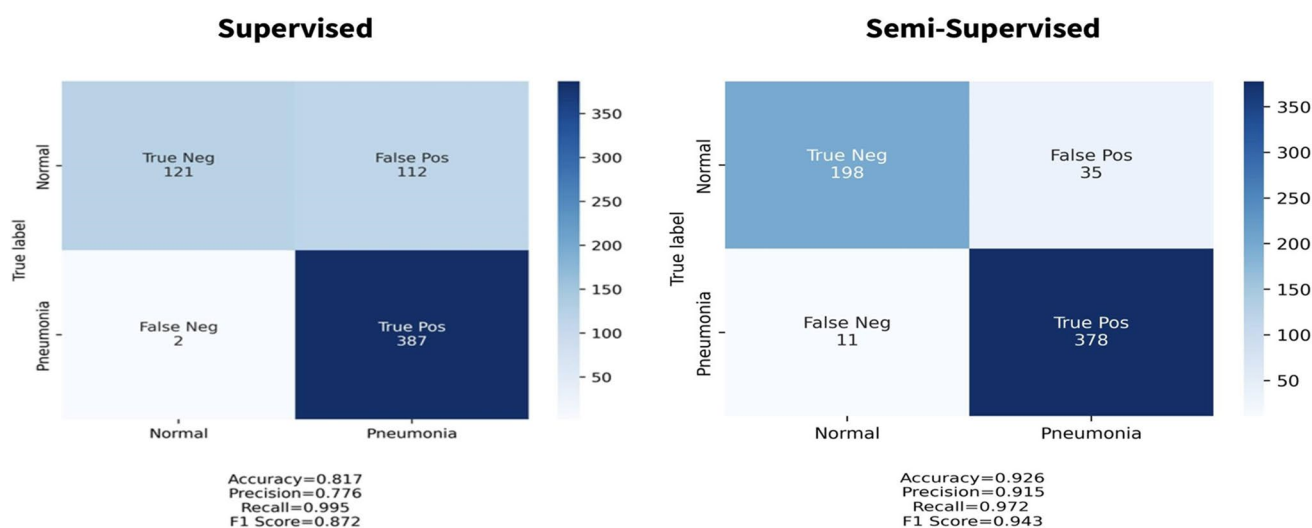


Fig. 15 Comparison of supervised learning and semi-supervised learning for Kaggle dataset

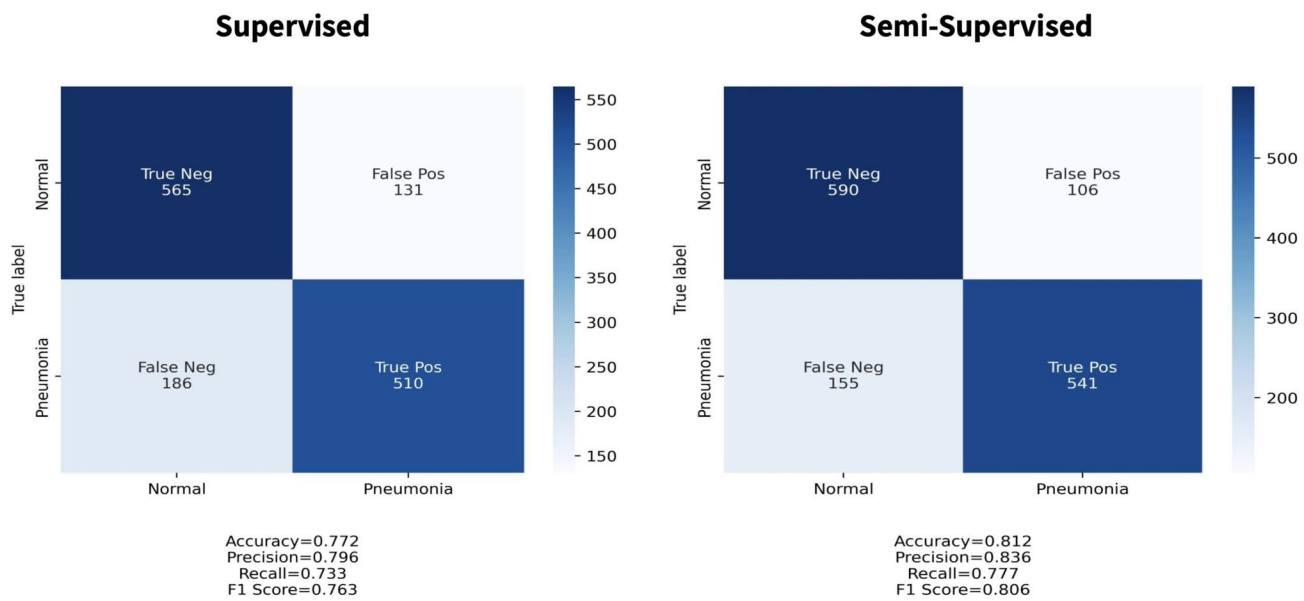


Fig. 16 Comparison of supervised learning and semi-supervised learning for CHEX dataset

ing refines past research that may have used all pseudo-labels indiscriminately. This method may improve learning from unlabeled data and filter for high-quality pseudo-labels, reducing training noise.

- **Benchmarking transfer learning and augmentation methods:** Multiple prominent architectures (VGG16, ResNet50, ViT) trained on different dataset sizes were used to evaluate transfer learning's pneumonia detection performance. Many earlier studies used transfer learning without a comparative context, leaving gaps in understanding how different architectures handle medical images of different quality and resolution. The results show that pre-trained ImageNet models can be a good starting point for smaller datasets, but they cannot capture subtle, domain-specific features.

Limitations of the Study

The study advances pneumonia detection using machine learning but has notable limitations:

- **Dependence on pre-trained models:** The use of models pre-trained on ImageNet (VGG16, ResNet50, ViT) may overlook medical imaging's unique, domain-specific features, and subtle pneumonia indicators, possibly affecting diagnostic accuracy.
- **Limited generalizability:** Although zero-shot and few-shot learning were explored, the study does not thoroughly assess model performance across diverse populations, which is crucial for clinical applicability.
- **Challenges with imbalanced data:** While class balancing was attempted, traditional methods may fall short in addressing real-world data imbalance. Proposed generative models like GANs are untested here, leaving gaps in evaluating this approach.
- **Basic semi-supervised learning:** The study uses the Mean Teacher algorithm and self-training with high-confidence pseudo-labeling. However, more advanced semi-supervised methods (e.g., FixMatch, MixMatch) could yield better results.
- **Model compression limitations:** Large models like VGG16 and ResNet50 are resource-intensive. While future work proposes pruning and knowledge distillation, these techniques were not tested, leaving practical deployment concerns.
- **Dependence on structured datasets:** The study used clean, well-organized datasets like Kaggle, BRAX, and CheXpert. However, in real-world scenarios, medical images often come with variations in quality, artifacts, and noise, which can impact model accuracy.
- **Insufficient exploration of uncertainty in medical labels:** The study acknowledges that CheXpert labels some images as "uncertain," using these as unlabeled data in semi-supervised learning. However, the model does not explicitly address or quantify the impact of label uncertainty on performance.

Conclusion

In our study, we explored several machine learning methods to tackle the challenge of training deep learning models for pneumonia detection. We evaluated their performance on multiple datasets of varying sizes, the Chest X-Ray Images (Pneumonia) dataset, the BRAX dataset, and the CheXpert dataset. Our key findings demonstrated that VGG16, when combined with class weights and data augmentation, achieved the highest accuracy of 91% on the Kaggle dataset, outperforming ResNet50 and Vision Transformers (ViT), which achieved accuracies of 86.7% and 88.5%, respectively. Transfer learning from the Chest X-Ray Images (Pneumonia) dataset to the smaller BRAX dataset yielded promising results in few-shot learning, reaching an accuracy of 82% with just 64 images from BRAX.

For the CheXpert dataset, incorporating semi-supervised learning with the Mean Teacher algorithm resulted in a 7.4% improvement in accuracy, achieving 83.5% compared to the 76.1% from models trained solely on labeled data. While zero-shot learning performed poorly on smaller datasets like BRAX, with an accuracy of only 58%, few-shot learning showed significant improvements as more data from the target dataset was included.

Moreover, using balanced weights and data augmentation significantly enhanced model performance on imbalanced datasets. For instance, on the Kaggle dataset, this approach improved accuracy by 10.3%, from 81 to 91%. These results underscore the effectiveness of transfer learning, data augmentation, and balanced weights in improving pneumonia detection accuracy, particularly in scenarios with imbalanced datasets or limited labeled data. Our study emphasizes the importance of choosing appropriate strategies based on dataset characteristics and highlights the potential of semi-supervised and transfer learning techniques in advancing biomedical image analysis.

Future Work

In future work, we plan to employ more sophisticated semi-supervised algorithms such as FixMatch and MixMatch to improve the performance of our pneumonia detection models on the CHEX dataset. To further optimize the models, we intend to apply Model Compression Algorithm, specifically considering pruning and label-based knowledge distillation. As a novel approach to addressing imbalanced datasets, we plan to implement generative models such as GAN and diffusion model for data augmentation. These techniques have the potential to significantly enhance the accuracy and robustness of the models and thus their clinical utility.

Data Availability No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Declarations

Institutional Review Board Statement Not applicable.

Informed Consent Not applicable.

Conflict of Interest The authors declare no competing interests.

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